

Example: Table on below gives the activities of a project, their predecessor(s), durations and costs.

a) Draw the network for the project.

b) Determine the critical path.

c) Construct the Time Schedule. $\rightarrow (HW)$.

d) Obtain programs with minimum costs.

$TD.$
 $\rightarrow TD_i < TD.$

Activities	Predecessor(s)	Durations		Costs	
		Normal	Crashed	Normal	Crashed
A	---	4	2	100	160
B	---	5	5	200	200
C	---	7	4	250	310
D	A	7	5	180	210
E	B	6	6	120	120
F	C	8	4	150	250
G	C	9	6	200	320
H	E, F	4	3	160	220
I	E, F	6	5	240	300
J	E, F, D	10	5	150	350
K	H, G	5	5	250	250

Slope
 $60/2 = 30.$

$60/3 = 20$

15

—

25

40

60

60

40

—

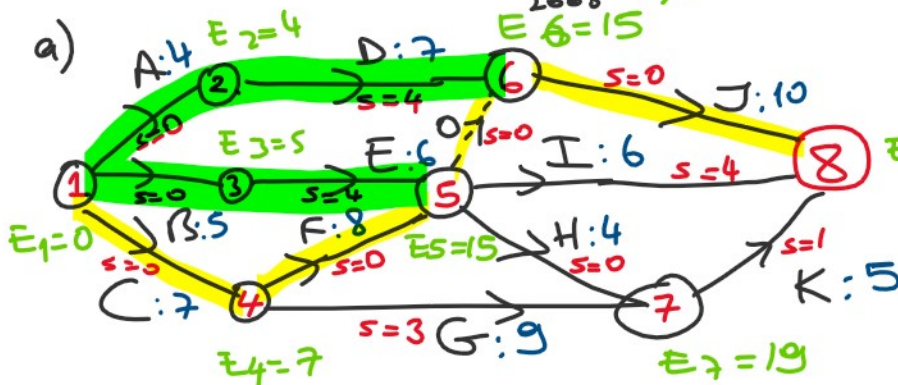
$E = \{E + D\}$
 $\max$

$L = \min \{L - D\}$

$FF \rightarrow s = E_j - E_i - D_{ij}$

$E_8 = \max(25, 21, 24) = 25.$

$E_i \xrightarrow{s=D_{ij}} E_j$



b)

Normal Program (PRO)

Critical Path = C F J

Total duration = 25.

Total cost = 2000.

$s = E_j - E_i - D_{ij}.$

By definition, the critical activities of a network must constitute an uninterrupted path that spans the entire network from start to finish.

Critical Act: C, F, J.
 20, 25, 40

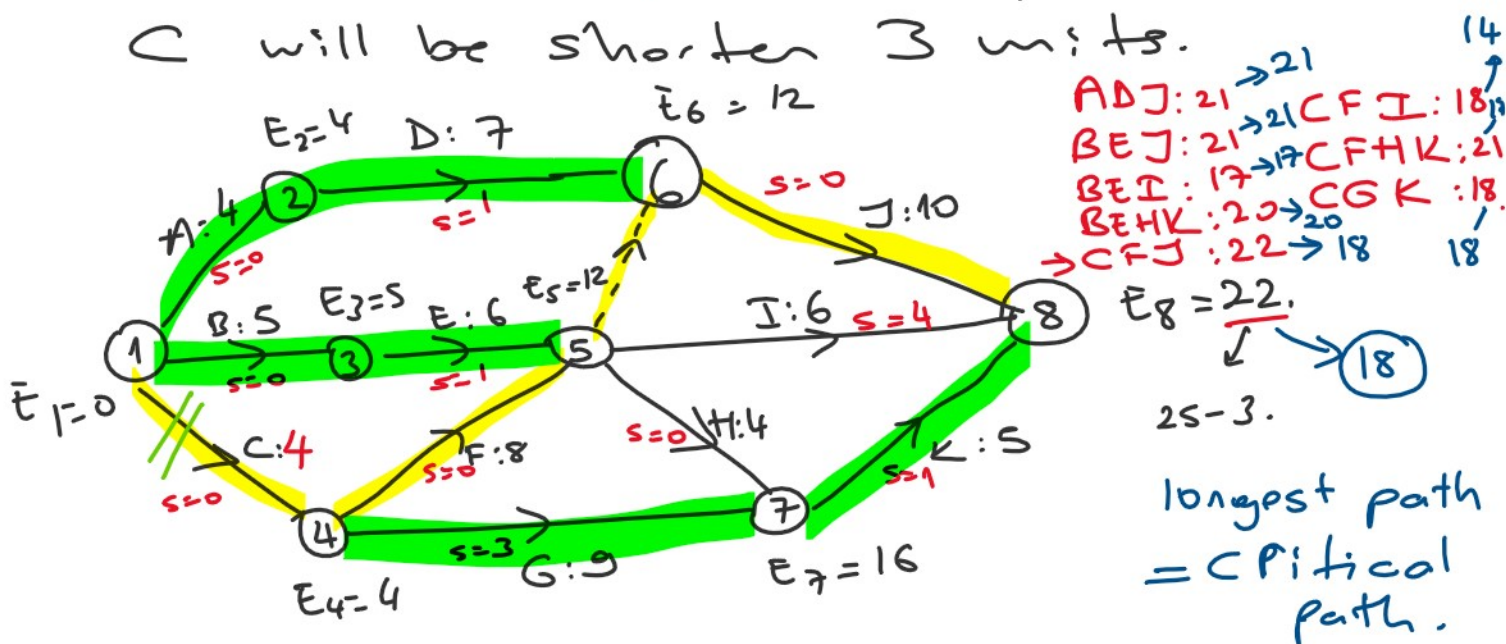
C act. will be crashed.

But how much? 1 unit, 2 units, or more.

Crashing Amount of Act. C = $\min \left\{ \underbrace{7-4}_C, \underbrace{0+4}_{AB}, \underbrace{0+4}_{BE} \right\} = 3$

Alternative paths.

C will be shorter 3 units.



PR1

C.P: CFJ.

T.D: 22.

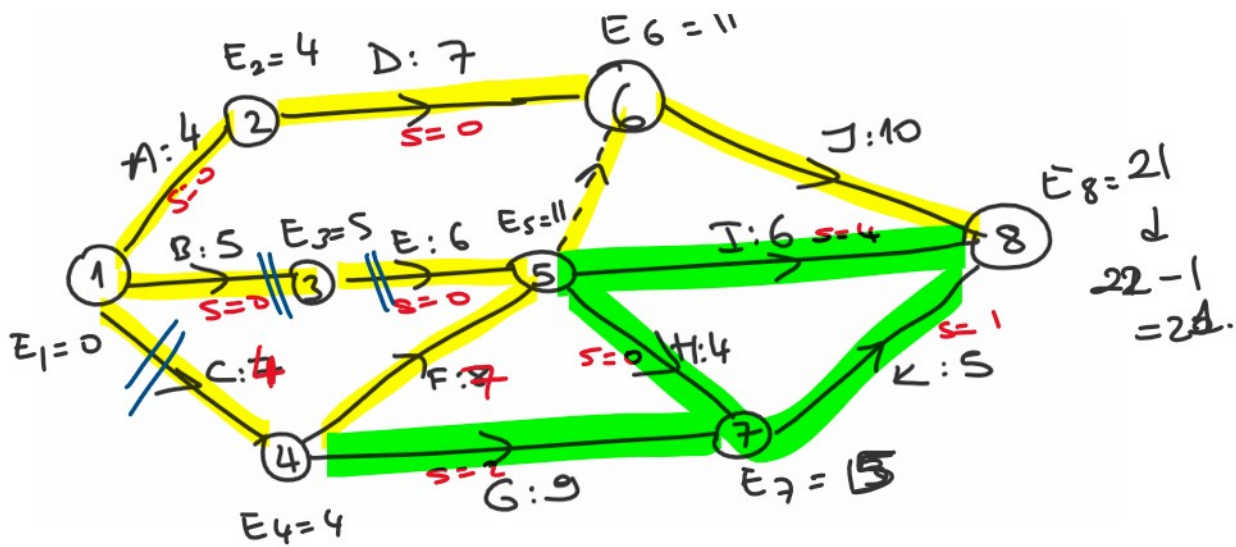
T.C: $2000 + 20 \cdot 3 = 2060$.

~~C~~ 20 F 25 J. 40.

Crashing Amount of Act. F = $\min \left\{ \underbrace{8-4}_F, \underbrace{0+1}_{AB}, \underbrace{0+1}_{BE}, \underbrace{3+1}_{GK} \right\} = 1$.

F will be crashed 1 unit.

$E_2 = 4$ D: 7 $E_6 = 11$



PR2

CP: ~~A~~, ~~D~~, ~~F~~, J

~~B~~, ~~C~~, ~~E~~, ~~G~~, ~~H~~, ~~I~~, ~~K~~

$J: 21 \rightarrow 21$

$A: 21 \rightarrow 20$

$B: 21 \rightarrow 21$

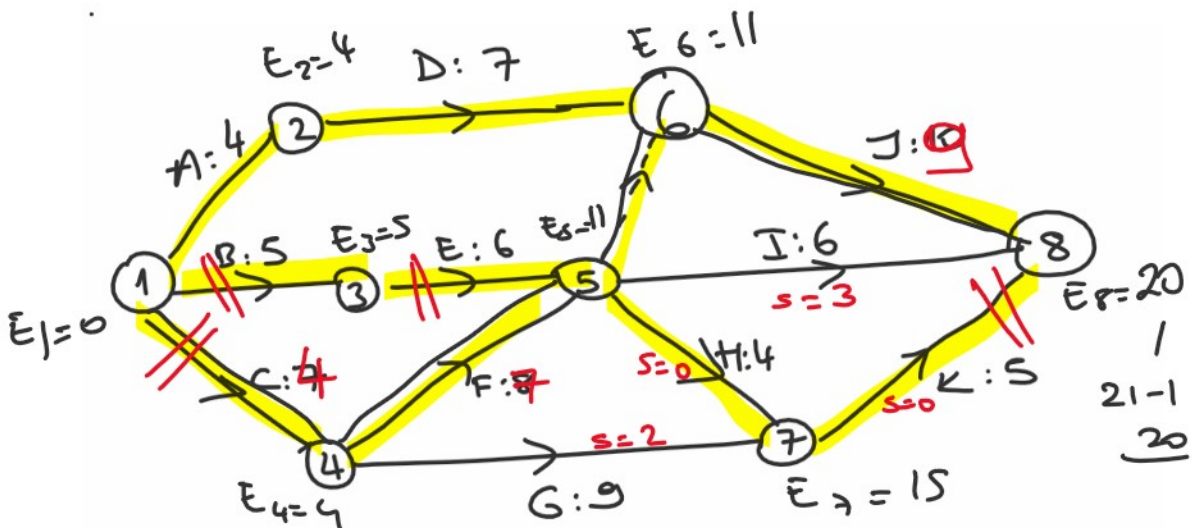
TD: 21

T.C: $2060 + 25 \cdot 1 = 2085$

Act J will be shorten 1 unit

C.A. of J = min $\left\{ 10-S, \begin{matrix} 4 \\ I \end{matrix}, \begin{matrix} 0+1 \\ H \end{matrix}, \begin{matrix} 2+1 \\ K \end{matrix} \right\}$

= 1



PR3

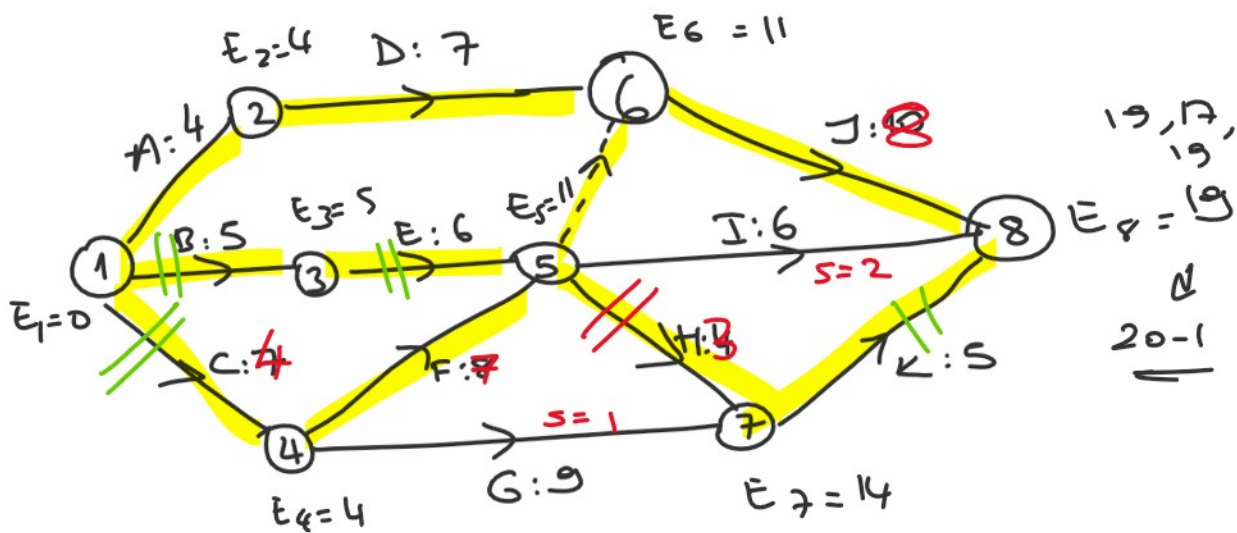
C.P: ~~XFJ~~:20 ✓
~~ADJ~~:20 ✓
~~BEJ~~:20 → J
~~BEHK~~:20 → H.
~~XFHK~~:20 ✓

J and H
 will
 be crashed.

$$T.D = 20$$

$$T.C = 2065 + 40 = 2125.$$

C.A. of Act J and H. = $\min \left\{ \frac{9-5}{J}, \frac{4-3}{H}, \frac{2}{I}, \frac{3}{G} \right\} = 1.$



PR4

C.P: ~~XFJ~~:19 → 18
~~ADJ~~:19 → 17
~~BEJ~~:19 → 17
~~BEHK~~:19 → 19
~~XFHK~~:19 → 17

19.

$$T.D = 19$$

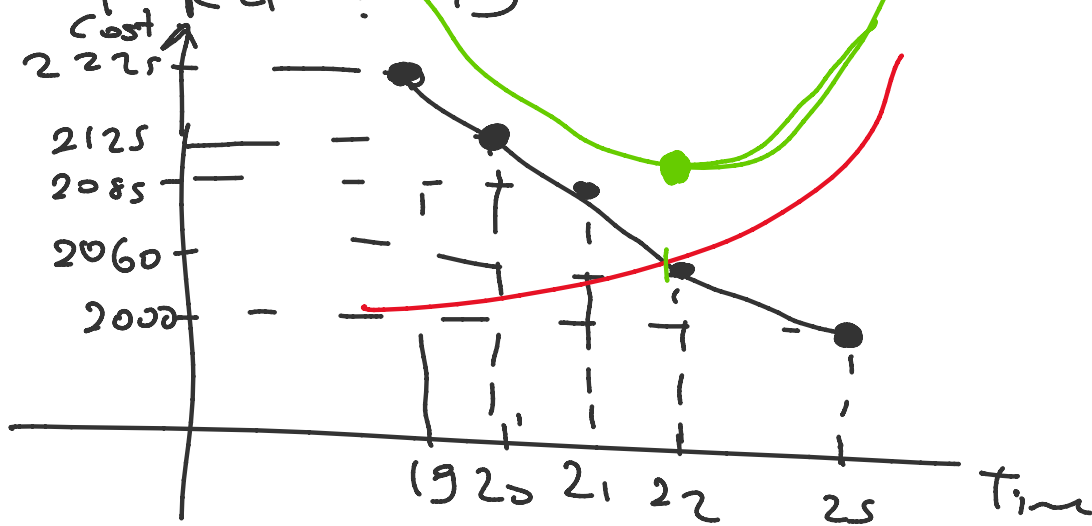
$$T.C = 2125 + 40.1 + 60.1 = 2225.$$

Since ~~BEHK~~ is reached its crashing limit, the crashing process is over.

TD TC

is over.

	TD	TC
RR0	25	2000
PR1	22	2060
PR2	21	2085
PR3	20	2125
PR4	19	2225



Example2: Table on below gives the activities of a project, their predecessor(s), durations and costs.

- Draw the network for the project.
- Determine the critical path.
- Obtain programs with minimum costs.

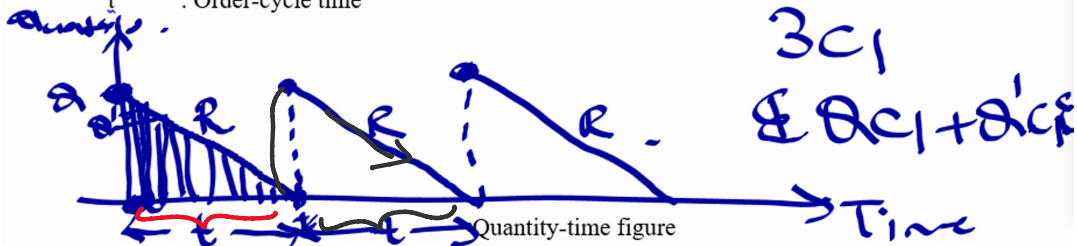
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D	A	7	5	200	260
E	B, C	4	4	120	120
F	B, C	7	7	150	150
G	C	5	3	180	250
H	D, F	8	6	200	300
I	E, G	9	6	100	160

Parameters of the model

- R : Demand rate (Consumption quantity per unit time)
- c_1 : Holding cost (\$/unit/unit time)
- c_2 : Cost of placing order (Ordering cost) (\$/order)

Variables of the model

- Q : Order quantity
- t : Order-cycle time



Total Inventory Cost = Holding cost + Cost of placing order

$$TIC = \frac{1}{2} \cdot t \cdot R \cdot c_1 + c_2 \rightarrow \min.$$

$$\text{Unit inventory Cost} = \frac{TIC}{Q} = \frac{1}{2} t c_1 + \frac{c_2}{Q}$$

$$UIC(t, Q) = \frac{1}{2} t c_1 + \frac{c_2}{Q}$$

$$Q = R \cdot t \Rightarrow \begin{matrix} 1^o) t = \frac{Q}{R} \\ 2^o) Q = R \cdot t \end{matrix}$$

$$UIC(Q) = \frac{1}{2} \frac{Q}{R} c_1 + \frac{c_2}{Q} \rightarrow \min.$$

$$\frac{d}{dQ} (UIC) = \frac{c_1}{2R} - \frac{c_2}{Q^2} = 0$$

$$\frac{c_1}{2R} = \frac{c_2}{Q^2}$$

$$Q^* = \sqrt{\frac{2Rc_2}{c_1}}$$

Optimal order quantity.

$$\frac{d^2 (UIC)}{dQ^2} = \frac{2c_2}{Q^3} > 0 \Rightarrow \min.$$

$$Q = R \cdot t \Rightarrow t = \frac{Q}{R}$$

$$t^* = \frac{Q^*}{R} = \sqrt{\frac{2C_2}{C_1 R}}$$

Optimal order cycle-time.

Example [Winston, Page 857]:

Braneast Airlines uses 500 taillights per year. Each time an order for taillights is placed, an ordering cost of \$5 is incurred. Each light costs 40\$ and the holding cost is 8 cent/light/year. Assume that demand occurs at a constant rate and shortages are not allowed.

- What is the optimal order quantity? $Q^* = ?$
- How many orders will be placed per year? $N = ?$
- How much time will elapse between the placement of orders? $\rightarrow t^* = ?$
- What is the total inventory cost? TIC per one cycle = ?
- What is the total inventory cost per year?

TIC per year?

$C_2 = \$5 / \text{order}$.

$R = 500 \text{ units / year}$.

$C_1 = \$0.08 \text{ unit/year}$.

$$Q^* = \sqrt{\frac{2RC_2}{C_1}} = \sqrt{\frac{2 \cdot 500 \cdot 5}{0.08}} = \frac{500}{2} = 250 \text{ units (a)}$$

$$t^* = \frac{250}{500} = \frac{1}{2} \text{ year} = 6 \text{ months (c)}$$

(b) How many orders will be placed per year? 2.

$$N = \frac{1 \text{ year}}{t_{\text{year}}} = \frac{1}{1/2} = 2.$$

T year

1/2

- d) What is the total inventory cost? **TIC per one cycle = ?**
e) What is the total inventory cost per year? **TIC per year?**



$$TIC = \frac{1}{2} \cdot Q \cdot t \cdot C_1 + C_2$$

$$= \frac{1}{2} \cdot 250 \cdot \frac{1}{2} \cdot (0.08) + 5$$

$$= 10 \$ / \text{cycle} \cdot (2)$$

$$\text{TIC}_{\text{per year}} = \underline{\underline{TIC}} \times \underline{\underline{N}} = 10 \times 2 = 20 \$ / \text{year}.$$